Gokul M

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Summary

AI/ML Engineer with internship experience building LLM-powered applications and production ML systems. Experienced in Python, SQL, FastAPI, and Docker with a strong focus on structured data querying, backend systems, and real-world deployment.

Experience

OneIndia Mar 2026 – May 2026

AI/ML Intern

- Built an IPL chatbot using an LLM-based SQL generation pipeline (User → LLM → SQLite → LLM → Response).
- Improved query accuracy by eliminating incorrect retrievals through prompt-driven structured querying.
- Developed backend logic for real-time query handling and response generation.
- Worked on additional ML-based production systems.

Techmaghi LLP, Kochi Aug 2025 – Feb 2026

Data Science Intern

- Built end-to-end ML pipelines including preprocessing, feature engineering, and evaluation.
- Developed resume ranking system using NLP (spaCy, FastText) with cosine similarity.
- Improved churn prediction recall from 86% to 90% using SVM and SMOTE on 200K+ dataset.
- Created Power BI dashboards for business insights.

Projects

Hybrid-Retrieval-Knowledge-Assistant-with-Citation-Verification | *Hybrid RAG, Qdrant, BM25*

- Built a production-oriented RAG pipeline using BGE dense retrieval, BM25, RRF fusion, and cross-encoder reranking, with fixed-size and structure-aware chunking experiments.
- Implemented grounded Gemini generation with strict context-only prompting, chunk-level citations, citation verification, confidence scoring, and missing-knowledge detection.
- Developed a 50-question evaluation framework comparing dense vs. hybrid retrieval; hybrid achieved 92.31% Hit@1, 99.15% Recall@5, 100% Hit@5, and 0.96 MRR.

Customer Churn Prediction API | *Python, FastAPI, Docker*

- Built ML system using SVM + SMOTE for class imbalance on 200K+ dataset.
- Improved recall from 86% to 90%.
- Developed FastAPI service and deployed using Docker.

Industrial AI PPE Detection System | *Python, YOLOv8, OpenCV, PyTorch*

- Fine-tuned a YOLOv8 Nano model using transfer learning for real-time detection of hard hats, safety vests, gloves, face masks, and safety boots.
- Developed a modular computer vision pipeline for live webcam and video-based PPE monitoring with real-time visualization and annotated video output.

Skills

Languages: Python, SQL
Machine Learning: Linear Regression, Logistic Regression, SVM, LightGBM, XGBoost, Feature Engineering, Model Evaluation, SMOTE, Classification Modeling, Time Series Forecasting, Backtesting, Computer Vision, Object Detection
Deep Learning: TensorFlow, Keras, LSTM, GRU, YOLOv8
LLM & NLP: Prompt Engineering,RAG, spaCy, FastText, BM25
Backend: FastAPI, REST APIs, Firebase Realtime Database, API Integration
Data: Pandas, NumPy, Scikit-learn, OpenCV, EDA, SQLite, yFinance
Deployment: Docker, Streamlit
Tools: Git, Power BI, Vector Databases(Qdrant), Roboflow

Education

Vimal Jyothi Engineering College 2021 – 2025

B.Tech in Artificial Intelligence and Data Science